

Juan Lorente

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Experience

RESEARCH INTERN

Graphics and Imaging Lab

Zaragoza, Spain

feb. - may. 2025

- Built Python pipelines to preprocess and normalize 360° video frames.
- Extracted saliency-based features to capture viewer attention hotspots.
- Trained, validated and tuned sequence-prediction models (Encoder/Decoder).
- Leveraged lab's high-performance servers to train compute-intensive models.

Education

BACHELOR'S DEGREE IN COMPUTER ENGINEERING

University of Zaragoza

Zaragoza, Spain

2024

Mention in Computing (Computer Vision - Graphics)

Bachelor Thesis '*Prediction of scanpaths in 360° videos*' with a DL method

MASTER'S DEGREE IN ROBOTICS, GRAPHICS & COMPUTER VISION

University of Zaragoza

Zaragoza, Spain

2026

GPA: 9.0 / 10

Relevant coursework: Autonomous Robots; Computer Vision; Machine Learning; Modeling and Simulation of Appearance; SLAM; Computational Imaging; Programming & Architecture of Computing Systems; Virtual Reality; Advanced Computational Imaging; Applications of Deep Learning

Projects

MASTER'S THESIS – Deep spatio-temporal scanpath prediction for 360° video

- *Description:* Designed and implemented a deep spatio-temporal framework that predicts temporal scanpath probability maps (tSPMs) for 360° video, integrates spherical convolutions and ConvLSTM temporal modeling, and uses a Gaussian-masked probabilistic sampler to generate coherent, human-like scanpaths.
- *Role:* Implemented the encoder-decoder architecture, training pipeline (KL-SoftDTW loss), heuristic sampling strategy, and extensive evaluation including cross-dataset testing and quantitative/qualitative analysis.
- *Technologies Used:* Python, PyTorch, TensorBoard, NVIDIA RTX A6000 GPU, Git.
- *Outcome:* Achieved superior performance versus saliency-based and state-of-the-art baselines on 360° eye-tracking tests; reproducible code and results published alongside the thesis.

3D WEB PORTFOLIO

- *Description:* A fully interactive, 3D portfolio site deployed on GitHub Pages that showcases my bio, education, other relevant sections and over 10 projects with links to code, documentation and live demos. (<https://juanlorenteguarnieri.github.io/portfolio/>)
- *Technologies Used:* React, Three.js, @react-three/fiber, GitHub Pages.
- *Outcome:* Live site with immersive 3D navigation, centralized access to all projects and credentials, facilitating easy sharing and ongoing updates.

Additional

Language Skills: Spanish (native), English (B2 – European Framework)

Technical Skills: C++, Python, PyTorch, Github, Linux, JavaScript.

Volunteer Experience: Summer Camp Counselor, 2023 – Planned and led children's activities; coordinated campsite setup and logistics.

Interests/Hobbies: Emerging Technology & AI, Piano performance (recreational), Psychology, Chess.